

Hibernate @OneToOne Mapping

MCQ Platform

Program

```
package com.nit.task;

import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;
import lombok.Data;

@Entity
@Table(name = "ans_value")
@Data
public class Answers {

    @Id
    @GeneratedValue(strategy = GenerationType.AUTO)
    @Column(name="ans_id")
    private int id;

    private String answers;
}

package com.nit.task;

import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.OneToOne;
import jakarta.persistence.Table;
import lombok.Data;

@Entity
@Table(name = "que_value")
@Data
```

```

public class Question {

    @Id
    @GeneratedValue(strategy = GenerationType.AUTO)
    @Column(name="que_id")
    private int id;

    private String quetions;
    @OneToOne
    private Answers answers;
}

```

```

package com.nit.task;

import org.hibernate.Session;
import org.hibernate.Transaction;

import com.nit.connectionProvider.ConnectionEstablisher;

public class MCQPlatform {
    public static void main(String[] args) {
        Session currentSession=ConnectionEstablisher
            .getConnection().openSession();

        Transaction transaction= currentSession.beginTransaction();

        Question que1=new Question();
        que1.setQuetions("What is <mapping> work?");

        Question que2=new Question();
        que2.setQuetions("What is hibernate?");

        Question que3=new Question();
        que3.setQuetions("What is use of @DependsOn ?");

        Answers ans1=new Answers();
        ans1.setAnswers("Hibernate is an ORM framework");

        Answers ans2=new Answers();
        ans2.setAnswers("It is use map the object to the database");

        Answers ans3=new Answers();
        ans3.setAnswers("execute the dependent class for one class");
    }
}

```

```

que1.setAnswers(ans2);
que2.setAnswers(ans1);
que3.setAnswers(ans3);

currentSession.persist(que1);
currentSession.persist(que2);
currentSession.persist(que3);

currentSession.persist(ans1);
currentSession.persist(ans2);
currentSession.persist(ans3);

System.out.println("Data added");

System.out.println("Fetch the data ");

Question reference = currentSession.getReference(Question.class,2);
System.out.println(reference.getQuestions());
System.out.println(reference.getAnswers().getAnswers());

transaction.commit();
currentSession.close();

}
}

```

What is One-to-One Mapping?

One-to-One mapping means **one object/entity is associated with exactly one object/entity of another entity.**

In this program:

One Question → One Answer

For example:

Question 1 → Answer 2
 Question 2 → Answer 1

Here, every **Question** has one **Answers** object.

What does @OneToOne do?

```
@OneToOne
private Answers answers;
```

@OneToOne tells Hibernate that the **Question** entity has a **one-to-one relationship with the Answers entity**.

Because of this mapping, Hibernate understands that:

Question → Answers

are related entities.

Therefore, we can directly access the answer through the question object:

```
reference.getAnswers().getAnswers()
```

Here:

```
reference
  ↓
Question object
  ↓
getAnswers()
  ↓
Answers object
  ↓
getAnswers()
  ↓
Actual answer
```

In This Program

This statement:

```
que1.setAnswers(ans2);
```

connects:

Question 1 → Answer 2

This statement:

```
que2.setAnswers(ans1);
```

connects:

```
Question 2 → Answer 1
```

This statement:

```
que3.setAnswers(ans3);
```

connects:

```
Question 3 → Answer 3
```

Then:

```
Question reference = currentSession.getReference(Question.class,2);
```

gets the **Question** object whose ID is **2**.

And:

```
System.out.println(reference.getQuestions());
```

gets the question.

Then:

```
System.out.println(reference.getAnswers().getAnswers());
```

uses the **@OneToOne** relationship to get the answer associated with that question.

Simple Definition

@OneToOne is used when one entity/object is associated with one entity/object of another type. It tells Hibernate about the one-to-one relationship between two entities.